THE RESIDUAL SPACE IS A FIELD SPECTRUM EXACTLY WHEN THE POINT IS REPRESENTED BY A MONOMORPHISM

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Editorial note. Counted span is the author's own proof environment, source lines 3316-3328, immediately following the statement at 3308-3314.

Editorial note. Two cross-chapter pointers (Morphisms of Spaces, Properties of Spaces) and two same-chapter line-indexed stubs (lemma-monomorphism-into-point, lemma-decent-points-monomorphism) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- The residual space is a monomorphism. definition of the residual space Z_x -> X used at proof line 3316 (Z_x -> X is a monomorphism by construction). The forward direction is immediate: if Z_x = Spec(k) the structural morphism represents x by a monomorphism.
- Fibres of monomorphisms towards a field. Morphisms of Spaces, Lemma spaces-morphisms-lemma-monomorphism-toward-field, cited at proof lines 3322-3324. A monomorphism of algebraic spaces towards Spec(k) whose source is nonempty is an isomorphism – used twice in the converse.
- Nonemptiness of the fibred product. Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian, cited at proof lines 3320-3321. Z_x x_X Spec(k) is nonempty, so the monomorphism towards Spec(k) is an isomorphism.
- Decent points are represented by monomorphisms. Lemma lemma-decent-points-monomorphism of this chapter, cited at proof line 3327. Supplies the final statement: decency of X gives the representing monomorphism for every point.

Source reference. Original lemma 03K4: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2565–2565. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06QY: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3184–3184. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06MG: focused-seeds/geometry/S9-decent-residual-space-monomorphism/ref-tags.txt, source lines 5–5. Navigation pointer

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-decent-residual-space-monomorphism/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $x \in |X|$. The residual space Z_x of X at x is isomorphic to the spectrum of a field if and only if x can be represented by a monomorphism $\mathrm{Spec}(k) \to X$ where k is a field. If X is decent, this holds for all $x \in |X|$.*

Proof. Since $Z_x \to X$ is a monomorphism, if $Z_x = \mathrm{Spec}(k)$ for some field k, then x is represented by the monomorphism $\mathrm{Spec}(k) = Z_x \to X$. Conversely, if $\mathrm{Spec}(k) \to X$ is a monomorphism which represents x, then $Z_x \times_X \mathrm{Spec}(k) \to \mathrm{Spec}(k)$ is a monomorphism whose source is nonempty by Properties of Spaces, Lemma 03H4. Hence $Z_x \times_X \mathrm{Spec}(k) = \mathrm{Spec}(k)$ by Morphisms of Spaces, Lemma 06MG. Hence we get a monomorphism $\mathrm{Spec}(k) \to Z_x$. This is an isomorphism by Lemma 06QY. The final statement follows from Lemma 03K4. □